

TAPACHULA, Mexico

WMO: 769030

Lat: 14.8872N

Lon: 92.2964W

Elev: 387

StdP: 14.49

Time Zone: -6.00 (MEX)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	68.2	69.7	60.8	80.7	76.7	62.8	86.6	76.8	28.9	89.6	21.0	79.1	3.2	40	0.305

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	18.2	96.0	79.0	94.8	78.9	93.7	78.9	83.1	91.5	82.2	91.0	81.5	90.3	8.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.9	162.6	89.5	79.7	156.6	88.4	78.9	152.4	87.7	47.4	91.5	46.3	91.3	45.4	90.6	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	14.3	10.7	DB	57.1	99.2	5.4	1.9	53.2	100.5	50.1	101.6	47.0	102.7	43.1	104.1	(4)
(5)			WB	57.2	85.0	2.7	1.9	55.3	86.3	53.7	87.5	52.2	88.5	50.3	89.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.4	80.8	82.4	83.9	84.9	84.0	82.2	82.3	82.4	81.5	81.6	81.7	80.7	(6)	
	DBStd	2.36	1.94	1.95	1.90	2.17	2.03	2.32	2.04	1.97	2.11	1.99	1.76	1.81	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD50	11810	954	907	1050	1046	1053	967	1003	1003	945	979	952	951	(10)	
	CDD65	6335	489	487	585	596	588	517	538	538	495	514	502	486	(11)	
	CDH74	63303	4542	5071	6579	6966	6450	4988	5287	5134	4278	4685	4903	4420	(12)	
	CDH80	27268	2035	2390	3151	3342	2841	1983	2225	2069	1546	1788	2000	1898	(13)	
(14) Wind	WSAvg	3.9	3.8	4.4	4.8	4.4	4.2	4.0	3.8	3.8	3.6	3.7	3.6	3.2	(14)	
(15) Precipitation	PrecAvg	92.8	0.4	0.1	0.9	3.5	10.2	14.5	12.2	13.5	16.6	12.5	3.8	0.3	(15)	
	PrecMax	121.6	2.8	1.3	4.9	13.1	23.0	31.8	28.7	23.5	31.1	27.4	14.6	1.9	(16)	
	PrecMin	64.8	0.0	0.0	0.0	0.0	1.2	4.0	2.9	6.8	9.1	0.8	0.0	0.0	(17)	
	PrecStd	14.4	0.7	0.2	1.4	2.6	4.6	5.7	5.0	4.0	5.8	5.7	3.9	0.4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.2	95.7	97.0	98.1	96.6	94.7	94.9	94.5	93.4	93.7	93.6	94.0	(19)	
		MCWB	76.9	77.1	78.5	79.4	79.4	81.6	81.3	81.5	81.0	80.6	80.0	77.6	(20)	
	2%	DB	92.6	94.2	95.4	96.1	94.4	92.5	93.2	92.8	91.5	91.8	91.9	92.0	(21)	
		MCWB	76.6	76.7	77.9	79.1	80.0	80.9	81.0	81.1	80.4	80.1	79.2	77.5	(22)	
	5%	DB	91.1	92.8	94.0	94.7	92.8	91.0	91.6	91.3	89.8	90.3	90.7	90.4	(23)	
		MCWB	76.3	76.4	77.6	79.0	80.1	80.4	80.4	80.4	79.7	79.8	78.8	77.5	(24)	
	10%	DB	89.6	91.4	92.5	93.0	91.1	89.4	89.9	89.5	88.1	88.6	89.2	88.8	(25)	
		MCWB	75.8	76.0	77.3	78.7	80.0	79.8	79.8	79.7	79.2	79.3	78.4	77.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.9	81.2	82.6	84.6	83.9	83.7	83.8	83.5	82.6	83.0	82.2	81.0	(27)	
		MCDB	90.5	91.6	92.8	92.5	91.6	91.9	91.7	91.9	91.1	90.4	90.8	89.1	(28)	
	2%	WB	78.9	79.3	80.4	81.9	82.3	81.9	82.0	82.0	81.2	81.6	80.9	79.8	(29)	
		MCDB	89.4	90.5	91.7	91.9	90.9	90.6	91.0	90.8	89.5	89.5	89.5	88.5	(30)	
	5%	WB	77.7	78.0	79.2	80.7	81.3	80.9	81.0	80.9	80.3	80.6	80.0	78.9	(31)	
		MCDB	88.3	89.6	90.8	91.4	90.2	89.4	89.9	89.5	88.3	88.4	88.4	87.9	(32)	
	10%	WB	76.7	76.9	78.2	79.6	80.4	80.1	80.1	80.0	79.4	79.7	79.2	78.0	(33)	
		MCDB	87.1	88.6	89.4	90.0	89.3	88.1	88.7	88.2	87.0	87.2	87.4	87.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	20.7	20.6	19.6	18.2	16.3	15.1	16.4	16.0	14.6	15.1	17.0	19.7	(35)	
		MCDBR	21.7	21.5	20.8	19.4	17.9	17.2	18.4	17.8	16.6	17.1	18.4	20.8	(36)	
	5% WB	MCWBR	10.4	9.5	8.7	7.8	7.8	8.2	9.5	9.3	8.3	8.6	9.4	10.6	(37)	
		MCWBR	20.9	20.6	19.7	18.5	16.9	16.4	17.6	16.8	15.9	16.0	17.5	19.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.397	0.423	0.496	0.597	0.639	0.491	0.496	0.484	0.482	0.441	0.416	0.399	(40)	
		taud	2.446	2.372	2.152	1.884	1.799	2.222	2.216	2.249	2.233	2.379	2.427	2.458	(41)	
	Ebn at Noon		285	281	263	236	223	256	255	261	263	272	276	281	(42)	
		Edh at Noon	35	39	50	65	70	45	46	45	45	38	35	34	(43)	
(44) All-Sky Solar Radiation	RadAvg		1693	1863	1977	1964	1718	1656	1844	1814	1628	1581	1645	1623	(44)	
	RadStd		44	68	65	84	115	99	118	105	115	125	68	50	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page